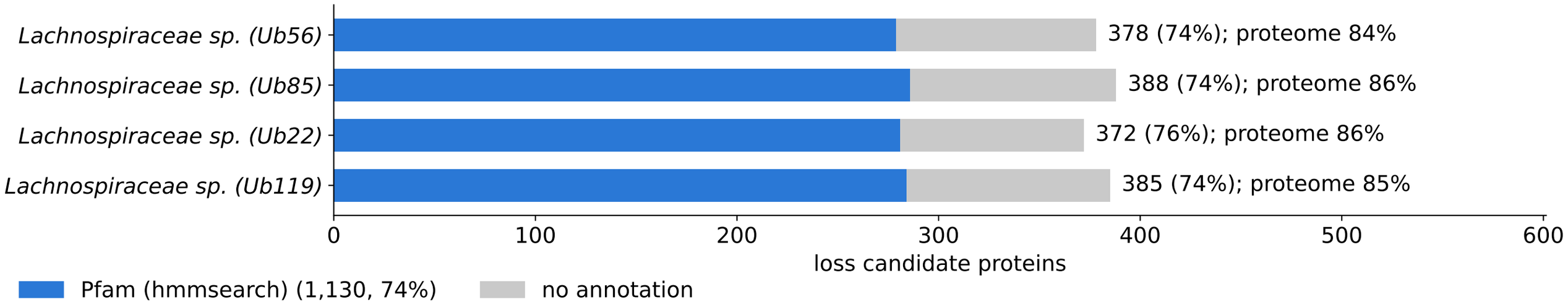
UHM MAG clade (4 OUT) vs Lachnospiraceae IN set (5 IN): enrichment of loss-direction candidates (present in >=0.75 of OUT, absent from all IN; diamond pairwise, NovInvenio run 2026-09-11)

A 1,523 candidates in 4 OUT species



B Pfam families enriched in candidates (primary; family units on both sides)

275 terms tested; 0 at FDR<0.05; annotated units: 454 cand. / 3011 all

No Pfam family passed FDR<0.05.

C GO: not available (Pfam-only annotation)

not available: no protein in this study has a term of this vocabulary; no test run

GO terms not available: no GO term annotation in this study.

Annotation: Pfam-A hmmsearch (HMMER 3.4, --cut_ga, Pfam 38.2) of all 9 proteomes (prodigal MAG proteins and RefSeq proteins); GO and InterPro not available. Universe: all proteins of the 4 OUT proteomes. Family units: loss-candidate families from the pipeline (mmseqs on loss_candidates.fa); background clustered with mmseqs (0.3 id, 0.8 cov, cov-mode 0). One-sided hypergeometric; BH within each vocabulary. Bars: FDR<0.05 and k>=3 candidate units. Loss candidates: 1523; annotated fraction 0.742.